

Vertical Carleman Completion of an Open-Set Ingham Problem on the Heisenberg Group

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Source question and status

Let $\mathbb{H}^n = \mathbb{C}^n \times \mathbb{R}$ be the Heisenberg group, let $T = \partial_t$ be its central derivative, and let $\mathcal{L}$ be its sublaplacian. Under the Schrödinger representation π_λ , $\mathcal{L}$ corresponds to the Hermite operator

$$H(\lambda) = -\Delta_{\mathbb{R}^n} + \lambda^2|x|^2.$$

Bagchi–Ganguly–Sarkar–Thangavelu [1] consider an integrable function f satisfying

$$\widehat{f}(\lambda)^* \widehat{f}(\lambda) \leq C \exp\left[-2\Theta(\sqrt{H(\lambda)})\sqrt{H(\lambda)}\right], \quad \lambda \neq 0, \quad (1)$$

where $\Theta : [0, \infty) \rightarrow [0, \infty)$ decreases to zero and

$$\int_1^\infty \frac{\Theta(r)}{r} dr = \infty. \quad (2)$$

Their Theorem 1.5 proves $f = 0$ if, for every $\lambda \neq 0$, the central Fourier fiber

$$f^\lambda(z) = \int_{\mathbb{R}} e^{i\lambda t} f(z, t) dt$$

vanishes on some open subset of $\mathbb{C}^n$. Remark 4.1 asks whether it is enough to assume instead that f vanishes on a nonempty open subset of $\mathbb{H}^n$.

proof. □

Remark 4.1. It would be interesting to see whether the conclusion of the Theorem 1.5 still holds true under the assumption that the function vanishes on a non-empty open subset of $\mathbb{H}^n$. A moment's thought staring at the above proof reveals that this can be achieved if we use an analogue of the Theorem 1.6 for the sublaplacian instead of special Hermite operators. But it turns out that proving an analogue of Theorem 1.6 is a very interesting and difficult open problem. We hope to revisit this in the near future.

The full question remains open. We prove it under an independent Denjoy–Carleman condition in the central direction.

Proof intuition

Ordinary open-set vanishing does not survive Fourier transformation in the central variable: a function can vanish for t in an interval while $f^\lambda(z)$ is nonzero for every λ . The added Carleman condition repairs exactly this loss. Pair $f(z, t)$ against an arbitrary horizontal test function $\eta(z)$. The resulting scalar function of t vanishes on an interval, and its even-derivative norms inherit the Carleman divergence from $T^{2m}f$. The one-dimensional Chernoff theorem makes the pairing vanish for every t . Varying η turns the local zero set into an entire vertical cylinder. The source's fiberwise theorem then applies without modification.

Theorem 1 (vertical Carleman subcase). *Assume (1) and (2). Suppose in addition that $f \in L^1(\mathbb{H}^n) \cap L^2(\mathbb{H}^n)$, that $T^{2m}f \in L^2(\mathbb{H}^n)$ for every $m \geq 1$, and that*

$$\sum_{m=1}^{\infty} \|T^{2m}f\|_{L^2(\mathbb{H}^n)}^{-1/(2m)} = \infty. \quad (3)$$

If f vanishes on a nonempty open subset of $\mathbb{H}^n$, then $f = 0$ almost everywhere.

Proof. Every nonempty open subset of $\mathbb{C}^n \times \mathbb{R}$ contains a product $U \times I$, where $U \subset \mathbb{C}^n$ and $I \subset \mathbb{R}$ are nonempty and open. Fix a nonzero $\eta \in C_c^\infty(U)$ and define, initially as an L^2 function,

$$g_\eta(t) = \int_{\mathbb{C}^n} f(z, t) \overline{\eta(z)} dz.$$

For every $m \geq 1$, distributional differentiation, Cauchy–Schwarz in z , and Fubini give

$$\|g_\eta^{(2m)}\|_{L^2(\mathbb{R})} \leq \|\eta\|_{L^2(\mathbb{C}^n)} \|T^{2m}f\|_{L^2(\mathbb{H}^n)}. \quad (4)$$

Thus g_η belongs to every even Sobolev space and has a smooth representative. Moreover, (4) and (3) imply

$$\sum_{m=1}^{\infty} \|g_\eta^{(2m)}\|_2^{-1/(2m)} = \infty;$$

the additional factor $\|\eta\|_2^{-1/(2m)}$ tends to one. Since $f = 0$ on $U \times I$, the function g_η vanishes on I , so it and all its derivatives vanish at any point of I . Chernoff's one-dimensional theorem [2], applied to the Laplacian $-d^2/dt^2$, yields $g_\eta \equiv 0$ on $\mathbb{R}$.

This holds for every $\eta \in C_c^\infty(U)$. Consequently $f = 0$ on $U \times \mathbb{R}$ in the distributional, hence almost-everywhere, sense. Fubini now shows that, for every central frequency for which the fiber is defined,

$$f^\lambda(z) = 0 \quad \text{for almost every } z \in U.$$

The fiberwise Ingham theorem, Theorem 1.5 of [1], applies under (1)–(2) and gives $f = 0$. $\square$

The hypothesis has a concrete analytic-vector sufficient condition.

Corollary 2 (central analytic vectors). *Under the other hypotheses of Theorem 1, condition (3) holds whenever*

$$e^{a|T|}f \in L^2(\mathbb{H}^n)$$

for some $a > 0$, where $|T|$ is defined by Fourier transformation in the central variable.

Proof. Spectral calculus and $\sup_{s \geq 0} s^{2m} e^{-as} = (2m/(ae))^{2m}$ give

$$\|T^{2m}f\|_2 \leq \left(\frac{2m}{ae}\right)^{2m} \|e^{a|T|}f\|_2.$$

After taking $-1/(2m)$ powers, the resulting series dominates a constant multiple of $\sum_m m^{-1}$. $\square$

Why this does not settle the full question

The added condition is not a formal consequence of the Hermite-only bound. The lowest Hermite eigenvalue gives $H(\lambda) \geq n|\lambda|$, so (1) controls central frequency only at the scale

$$\exp[-c\sqrt{|\lambda|}\Theta(c\sqrt{|\lambda|})].$$

Even if Θ were replaced by a positive constant, the associated $2m$ -th central derivative norm has a root of quadratic order in m , and the reciprocal Carleman series need not diverge. The actual hypothesis $\Theta(r) \rightarrow 0$ is weaker still. Thus the proof isolates an independent vertical quasi-analyticity mechanism; it does not hide a proof of the general sublaplacian Chernoff theorem.

Verification and novelty status

The proof uses only Cauchy–Schwarz/Fubini, the classical one-dimensional Chernoff theorem, and the source’s proved fiberwise Theorem 1.5. Exact-title, author, sublaplacian-Chernoff, and open-set searches were performed in the local parsed arXiv corpus and in arXiv-indexed web results through 2026-08-11. The closest sources checked were arXiv:2009.14230, arXiv:2106.02704, and arXiv:2011.09940; no later answer to Remark 4.1 was found. Novelty confidence for this modest added-Carleman subcase is medium, not high. Human review should focus on the passage from scalar test pairings to the common vertical cylinder and on the precise regularity convention in the source’s Theorem 1.5.

References

- [1] S. Bagchi, P. Ganguly, J. Sarkar, and S. Thangavelu, *An analogue of Ingham’s theorem on the Heisenberg group*, arXiv:2204.10017 (2022), especially Theorem 1.5 and Remark 4.1.
- [2] P. R. Chernoff, *Quasi-analytic vectors and quasi-analytic functions*, Bull. Amer. Math. Soc. 81 (1975), 637–646. The one-dimensional case is also stated as Theorem 1.6 in [1].